THE PRIME EXAMINATIONS 2022

P.7 PRE – PLE TRIAL SET VI

MATHEMATICS (Abridged Curriculum)

Time allowed 2 hours 30 minutes

INDEX NO:	Random No.						Personal No.		


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Candidate's Name:.....

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READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has **two** sections: **A** and **B** Section **A** has **20** questions and section **B** has **12** questions.
3. Answer **all** questions. **All** the working for both sections **A** and **B** must be shown in the spaces provided.
4. **All** working **must** be done using a **blue** or **black** ball point pen or ink. Any work done in pencil other than graphs and diagrams will **not** be marked.
5. **No calculators** are allowed in the examination room.
6. Unnecessary **changes** in your work and handwriting that cannot easily be read may lead to **loss of marks**.
7. Do **not** fill anything in the table indicated "**For Examiners' use only**", and those boxes inside the question paper.

FOR EXAMINERS' USE ONLY		
QUESTION NUMBER	MARKS ATTAINED	INITIALS
1 - 5		
6 - 10		
11 - 15		
16 - 20		
21 - 22		
23 - 24		
25 - 26		
27 - 28		
29 - 30		
31 - 32		
TOTAL		

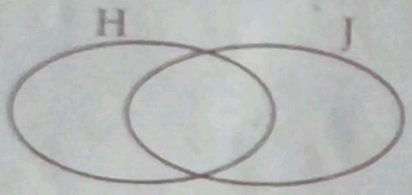
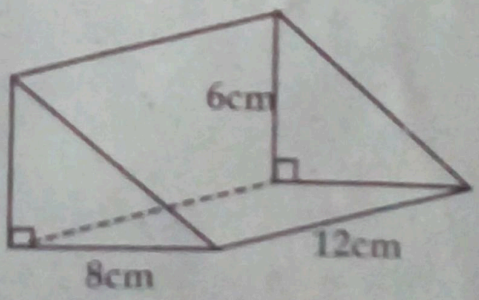
APPROVED:
Consultant
Mathematics Department (PEC)

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Turn Over

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Section A (40 Marks)

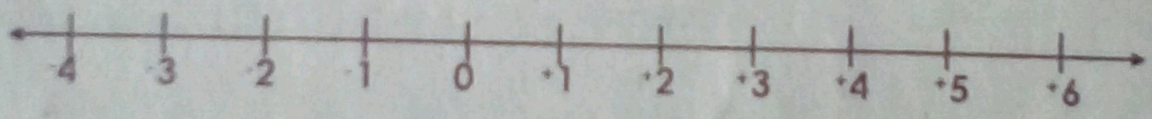
1	Workout: $24 \div 8$	2	Write in figures, "forty six thousand sixty one"
3	Given that set H = {2, 3, 5, 7, 11} and set J = {1, 2, 3, 4, 6, 12}. Show the elements of the two sets on Venn diagram. 	4	Find the mean of $(x + 4)$, 21, $2x$ and $(x + 7)$.
5	The clock face below shows time in the evening. Write the time shown in 24-hour clock. 	6	Simplify: $3(2 - y) + 6y$.
7	A motorcycle consumes $\frac{1}{4}$ litres of petrol for every 6km. How many litres of petrol can it use to cover 48km?	8	Find the volume of the triangular prism below. 

9 Using a protractor, a ruler and a sharp pencil. Draw an angle of 60^0 in the space below.

10 Find the next number in the sequence.
2, 3, 7, 16, 32, _____

11 Workout: (75 x 38) + (25 x 38)

12 Workout: $^+3 + {}^-7$ on the number line below.



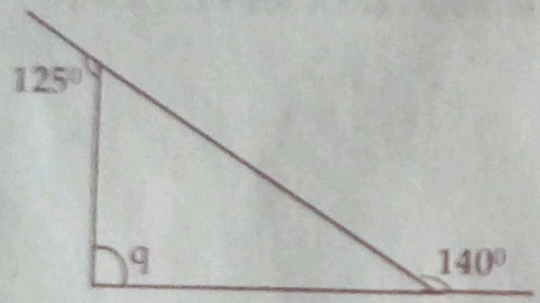
13 Round off 6789 to the nearest hundreds.

14 Ajidiru drove for $1\frac{1}{2}$ hrs from town R to town T. Town R is 87km from town T. At what speed was Ajidiru travelling?

15 Draw tallies to represent 23 pupils.

16 Given the exchange rates below.
KSh1 = Ush.36
US$1 = Ush.3500.
Convert USD 180 to Kenya shillings.

17 Find the value of q in degrees.



18 Increase sh.45000 by $12\frac{1}{3}\%$.

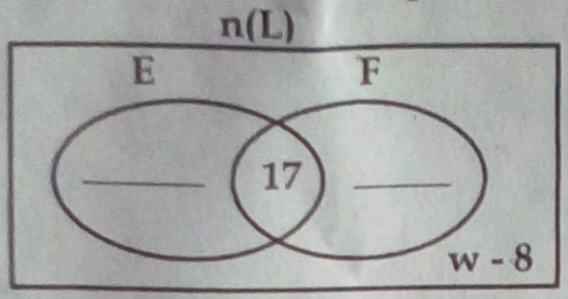
19 Solve: $3n + 2 < 14$

20 Cups of 500ml were used to fill a ten litre jerrycan with milk. Find the number of full cups which were used.

Section B (60 Marks)

21 **In a certain class, all pupils speak Luganda. Given that (2w + 3) pupils speak Luganda and English (E) only, 28 pupils speak Luganda and French. (F) only, 17 speak all the three languages while (w - 8) pupils speak Luganda only.**

(a) Complete the Venn diagram below. *(2marks)*



(b) If **60 pupils** speak Luganda and English, find the value of w. *(2marks)*

(c) Find the total number of pupils in the class. *(1mark)*

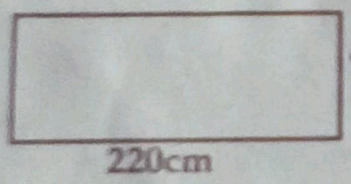
22 (a) Express 0.2666..... as a vulgar fraction. *(2marks)*

(b) Simplify: $\frac{3}{4} \times \frac{4}{9} \div \frac{4}{5} + 1\frac{1}{2}$ *(3marks)*

23 Find the sum of the value of 2 and the value of 3 in the number 1302_{four}. *(3marks)*

(b) Workout: $23_{five} \times 14_{five}$ *(2marks)*

24 **The rectangular sheet of metal below was to be curved into a cylindrical water tank.**



(a) Find the diameter of the cylindrical tank. **(Take $\pi = \frac{22}{7}$)** *(2marks)*

(b) How much water in litres can the cylindrical tank hold when full? *(3marks)*

A pupil bought the following items from the market.

3kg of sugar at sh.4000 a kg.
500g of rice at sh.3000 per kg.
5 kg of wheat flour at sh.7200 for every 2kg.

(a) How much money did the pupil pay altogether? *(4marks)*

(b) If the pupil was given a balance of sh.2500, how much money did the pupil have at first? *(1mark)*

26 (a) Using a ruler and a pair of compasses only, construct a parallelogram WXYZ where $\overline{WX}$= 8cm, angle WXY = 45^0 and $\overline{WZ}$= 5cm. *(4marks)*

(b) Measure the length of diagonal WY. *(1mark)*

27 The table below shows the masses of boys recorded during the first week of the term.

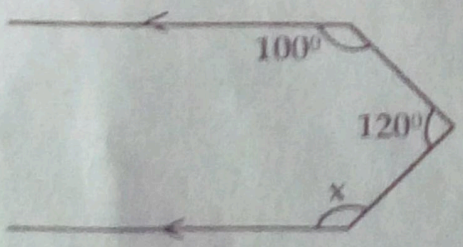
Mass(kg)	55	52	45	x	48
No. of boys	4	1	2	2	1

(a) If their mean mass is 49kg, find x. *(3marks)*

(b) Find the median mass. *(2marks)*

28 (a) The interior angle sum of a regular polygon is 540^0. How many sides has the polygon? *(3marks)*

(b) In the figure below, find the size of angle x. *(2marks)*



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29 Kana, Sande and Juma shared books in the ration **4:5:6** respectively. If Kana and Sande both got **36 books**,

(a) how many books did they share altogether? *(3marks)*

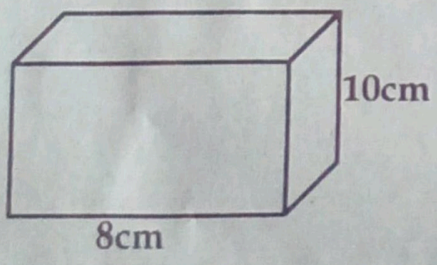
(b) Find the number of books Juma got. *(2marks)*

30 Madra is **56** years old and his son is **8** years old.

(a) After how many years will Madra be thrice as old as his son? *(3marks)*

(b) How old will the son be then? *(2marks)*

31 **The volume of the cuboid below is 480cm³.**



(a) Find the width of the cuboid. *(3marks)*

(b) Work out the total surface area of the cuboid. *(2marks)*

32 A taxi travelling at a steady speed of **60km/hr** took **3 hours** to travel from town **A** to town **B**. It rested for **30minutes** at **B** and returned to **town A** at a speed of **72km/hr**. Calculate the average speed of the taxi for the whole journey. *(5marks)*

Hidden Tip For Excellence in Mathematics.



- Passionately engage in study groups especially for those areas/concepts that are challenging to you.

- During these discussions, give and as well listen to other members' views to get the most appropriate answers.